1 Fast Fourier Transform (FFT)

The **Fast Fourier Transform (FFT)** is an efficient algorithm used to compute the Discrete Fourier Transform (DFT) and its inverse. The FFT reduces the computational complexity from $O(N^2)$ to $O(N \log N)$ by exploiting the symmetries in the DFT.

1.1 DFT Definition

For a given sequence of N complex numbers $x_0, x_1, \ldots, x_{N-1}$, the Discrete Fourier Transform is defined as:

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi kn/N}, \quad \text{for } k = 0, 1, \ldots, N-1$$

Where: - X_k represents the frequency-domain coefficients. - $e^{-i2\pi kn/N}$ are the complex roots of unity, which correspond to the periodic nature of the signal. - x_n is the time-domain input.

1.2 Complex Roots of Unity

The N-th complex roots of unity are the solutions to the equation:

$$z^N = 1$$

These roots are used by the FFT to divide the problem into smaller subproblems, and they form the basis of the efficient recursion in the algorithm.

1.3 Divide-and-Conquer Approach

The FFT applies a divide-and-conquer strategy by recursively breaking down the DFT of size N into two DFTs of size $N/2$. This uses the fact that:

$$X_k = \sum_{n=0}^{N/2-1} x_{2n} e^{-i2\pi kn/N} + e^{-i2\pi k/N} \sum_{n=0}^{N/2-1} x_{2n+1} e^{-i2\pi kn/N}$$

This approach allows the FFT to halve the number of operations at each recursive step, ultimately reducing the complexity.

1.4 Polynomial Interpretation

The FFT can be seen as evaluating a polynomial of degree $N-1$ at N distinct points (the N-th roots of unity). Given the polynomial:

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{N-1} x^{N-1}$$

FFT computes the values $P(\omega^k)$ for $k = 0, 1, \ldots, N-1$, where $\omega = e^{-i2\pi/N}$ is the principal N-th root of unity.

1.5 Applications

FFT is widely used in: - **Signal Processing**: Converting time-domain signals to frequency-domain for analysis. - **Image Processing**: Used in compression algorithms and filtering. - **Data Analysis**: Finding frequency components in data sets.

1.6 Real and Imaginary Components

When processing real-world signals (e.g., voltage samples), the input to FFT can be represented as real/imaginary pairs. If the input is real, the imaginary part is set to zero:

$$\text{Input Pair: } (x_{\text{real}}, 0)$$

After the FFT, the real and imaginary components of each frequency are populated, giving the frequency-domain representation of the signal.